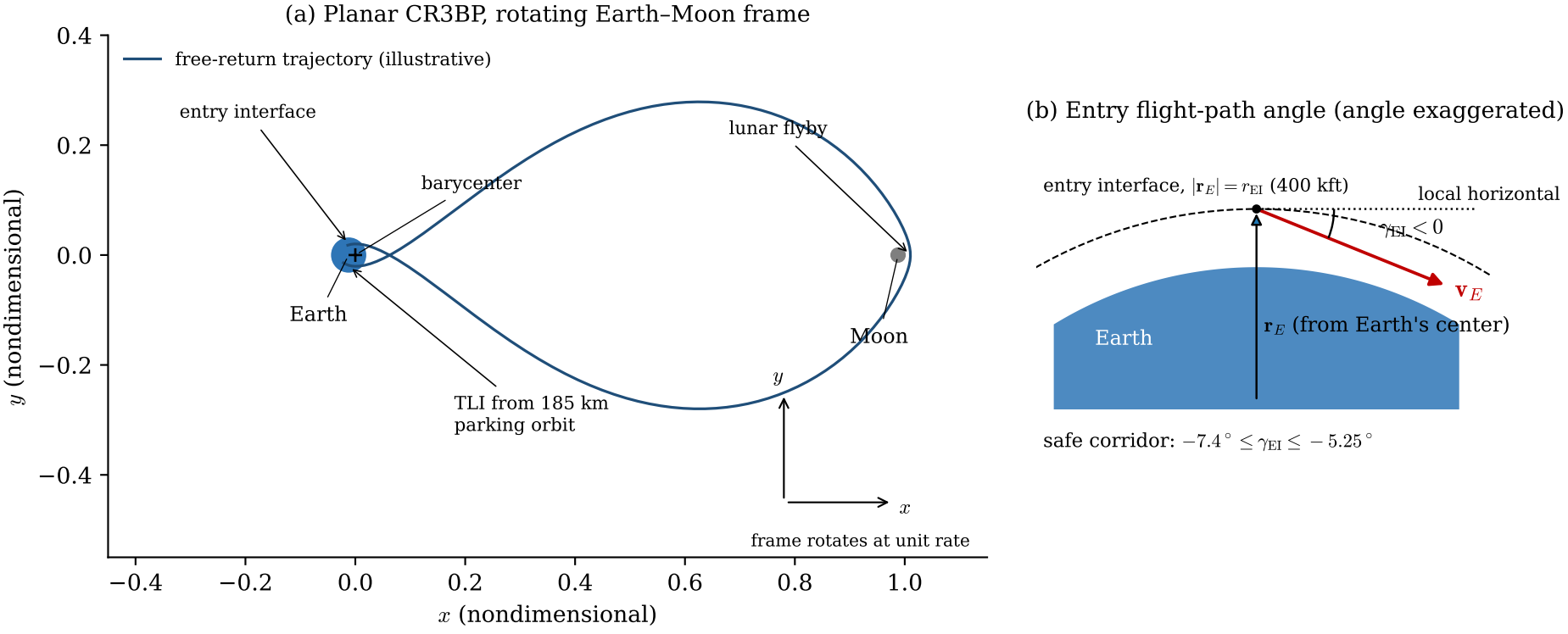
(a) Planar CR3BP, rotating Earth–Moon frame
free-return trajectory (illustrative)
entry interface
barycenter
lunar flyby
Earth
Moon
TLI from 185 km
parking orbit
y
x
frame rotates at unit rate
y (nondimensional)
x (nondimensional)
0.4
0.2
0.0
−0.2
−0.4
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
(b) Entry flight-path angle (angle exaggerated)
entry interface, $|\mathbf{r}_E| = r_{\mathrm{EI}}$ (400 kft)
local horizontal
$\gamma_{\mathrm{EI}} < 0$
$\mathbf{v}_E$
Earth
$\mathbf{r}_E$ (from Earth's center)
safe corridor: $-7.4^\circ \le \gamma_{\mathrm{EI}} \le -5.25^\circ$